



Conti Ransomware Peel-Chain Case Study

A Bitcoin tracing case study showing a peel-chain phase followed by consolidation.

Case focus

Peel-chain laundering is a way criminals try to hide ownership of cryptocurrency funds. Rather than moving one large balance directly, smaller amounts are split off over multiple transactions to make the linked addresses look unrelated.

This case study starts from the seed address `13XELohdTXGbgHpwRrNHMiKrjA7a9wM2z` and follows a high-value transaction output forward through the blockchain. The trace follows the continuing balance from one spent output to the next, then calculates the value separated from that continuing flow at each hop.

This case shows a clear peel-chain pattern, where most of the value continues forward while smaller amounts are peeled away across successive transactions. The traced path later ends when the remaining 20 BTC is merged into a much larger consolidation transaction, showing how laundering behaviour can shift from fragmentation to aggregation.

How the trace works

Transaction-by-transaction tracing

The app does not group all outputs by address. It follows a specific transaction output, checks whether that output was spent and then moves into the transaction that spent it. The largest continuing output in that transaction becomes the next step in the chain.

The peeled-off value is calculated as the difference between the balance entering the hop and the continuing balance leaving the hop.

Starting flow	Displayed hops	Total peeled off	Largest peel	Final displayed flow
102.5892 BTC	6	82.5892 BTC	24.2699 BTC	20.0000 BTC
Consolidation inputs	Consolidated output		Consolidation fee	
4	13449.6075 BTC		0.000408 BTC	

What the trace shows

The main flow begins at approximately 102.5892 BTC and is followed through successive spent outputs. Across the displayed path, 82.5892 BTC is separated from the continuing flow.

The important feature is the repeated forward movement: each hop spends the current output, creates a new continuing output and separates value away from the main path. The traced path then ends when the remaining 20 BTC output is swept into a larger consolidation transaction.

1. Peel-chain phase

The graph follows the continuing balance across successive spent outputs. The pink nodes show value separated from the main flow at each hop.

Conti peel-chain tracing path



2. Peel-chain evidence table

Step	Date Created	Continuing Address	Continuing BTC	Spending Transaction	Next Continuing E
1	22 Oct 2020	147iJKDhKwrdBTyUGamy2qi9yDjmbuLXKi	102.5892	8d29f54cea865adeb8995509c5313508f6cad4ac04de52751a0773a7acbd7077	91.1908
2	22 Oct 2020	13eChZ48XFNsibX9bduiX4entnaf1St3Re	91.1908	d82cd8d867f9b12e99e0a2c7f887e0a2963d47bee4c89fcd764c7b38300ff39e	68.3944
3	22 Oct 2020	13DT9VpeKzdVvpcRFRjpiVVCywUca7VL3o	68.3944	f02afee2e896e469936c73ee66daace5fea8679bab384d84866c0d3877ce75a	64.4219
4	23 Oct 2020	19G19JkrqSFRXJT3hG3czcRPN923sukV4	64.4219	62b3edb21fa397df8e089321a9ee4c2afef400d96d07c3196ddc5ac262690215	40.152
5	23 Oct 2020	19Cvj7i7pAqqj7yBuZg5RgawtDXJKf4guR	40.152	bfa7a9f0a12143dc01790a2e5913a4f758316714a0b56f6a8e4fe37c32f2f94e	30.1519
6	15 May 2022	13pN53hieQ3A28zq7ksmSWyR6rekf4Jtdg	30.1519	ee710e96f29d702a68a371274c97e24358cdc5b7439ed581fb6f5613826fc77b	
7	15 May 2022	3GdhZAUdk7BpVskTQbUmsRxmz3iUQDRy9L	20	253a0fc9dd7746fc1a395425d45f3ed23f98861e6e09465428501c4711e44d82	

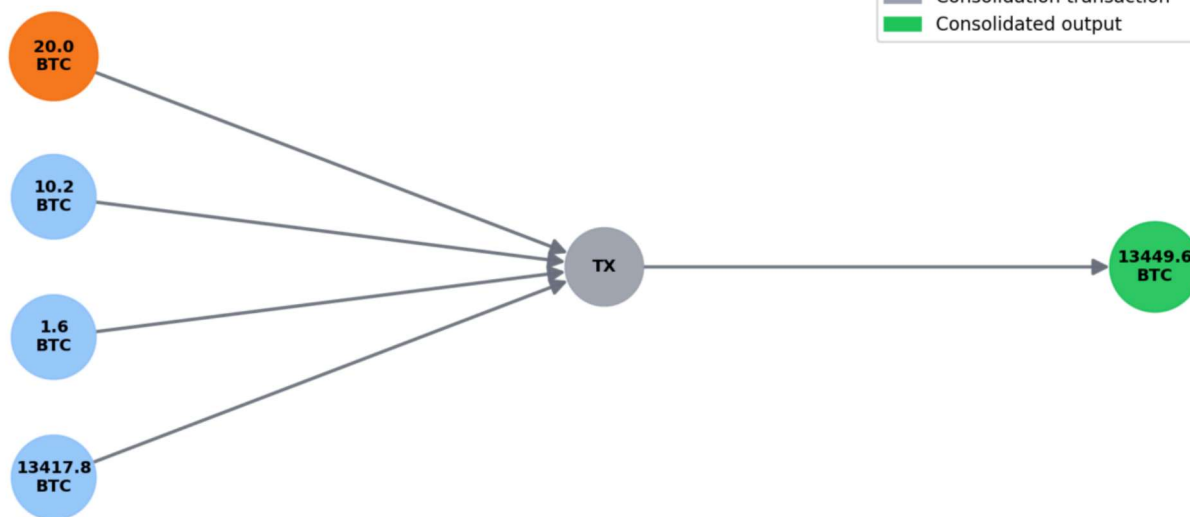
Show peeled-off output details

Step	Spending Transaction	Output Index	Output Address
1	8d29f54cea865adeb8995509c5313508f6cad4ac04de52751a0773a7acbd7077	0	bc1qw6uz7ye603spgu370tuhv3m6ff94hrry...
2	d82cd8d867f9b12e99e0a2c7f887e0a2963d47bee4c89fcd764c7b38300ff39e	0	bc1qe9je7694shnd9qr62vwgce9rcg0kpl0ja...
3	f02afee2e896e469936c73ee66daace5fea8679bab384d84866c0d3877ce75a	0	32fevJNUvq8VfLv7HZ8Ftx4QWNCJy7faNv...
4	62b3edb21fa397df8e089321a9ee4c2afef400d96d07c3196ddc5ac262690215	0	12aQfCc93wJmBqi7bNydiJ8kepxKhZuj7q...
5	bfa7a9f0a12143dc01790a2e5913a4f758316714a0b56f6a8e4fe37c32f2f94e	0	3GdhZAUdk7BpVskTQbUmsRxmz3iUQDRy9L...
6	ee710e96f29d702a68a371274c97e24358cdc5b7439ed581fb6f5613826fc77b	0	1KRXTYbRdDhoPrUnqtbaAMFhtGb9R7deAZ...

3. Consolidation phase

After the peel-chain sequence, the final traced output of 20.0000 BTC is spent with other inputs in a consolidation transaction. The transaction combines 4 inputs into 1 output, with approximately 13449.6075 BTC leaving as the consolidated output.

- Final traced peel-chain output (20 BTC)
- Other consolidation inputs
- Consolidation transaction
- Consolidated output



4. Transaction summary

Point	Finding	BTC
Starting flow	The traced path begins from a high-value transaction output.	102.589
Displayed hops	The app follows the continuing output transaction-by-transaction.	6
Total value separated	This is the total difference between the continuing balance at each displayed hop.	82.5892
Final displayed balance	This is the continuing balance at the end of the displayed path.	20.0000

Blockchain tracing shows transaction movement. The labels in this case describe transaction structure: starting flow, continuing flow and value peeled off. They do not identify a real-world person.

- ▼ Show references and data sources

Kappos, G., Yousaf, H., Stütz, R., Rollet, S., Haslhofer, B. and Meiklejohn, S. (2022) 'How to Peel a Million: Validating and Expanding Bitcoin Clusters', *31st USENIX Security Symposium*.